C Hackathon Folder Structure

GROUP C11

Location: Kolkata

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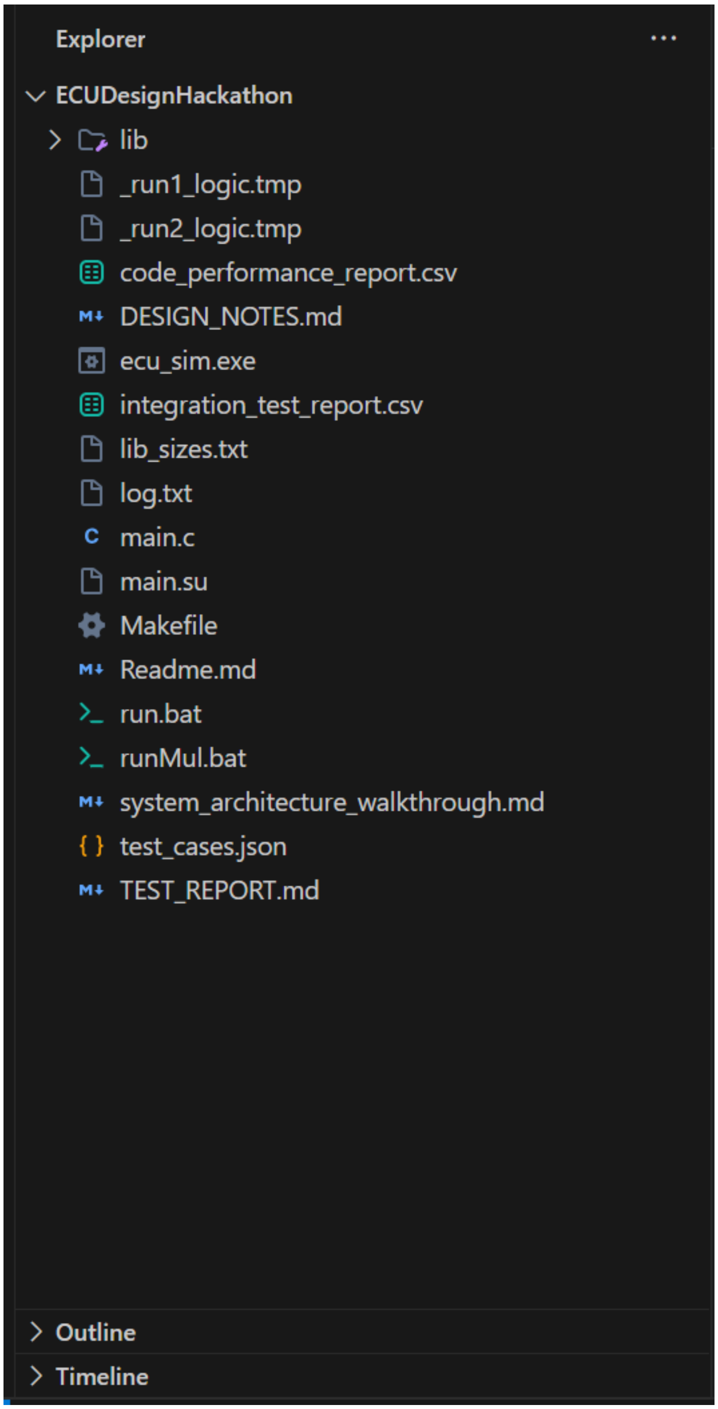
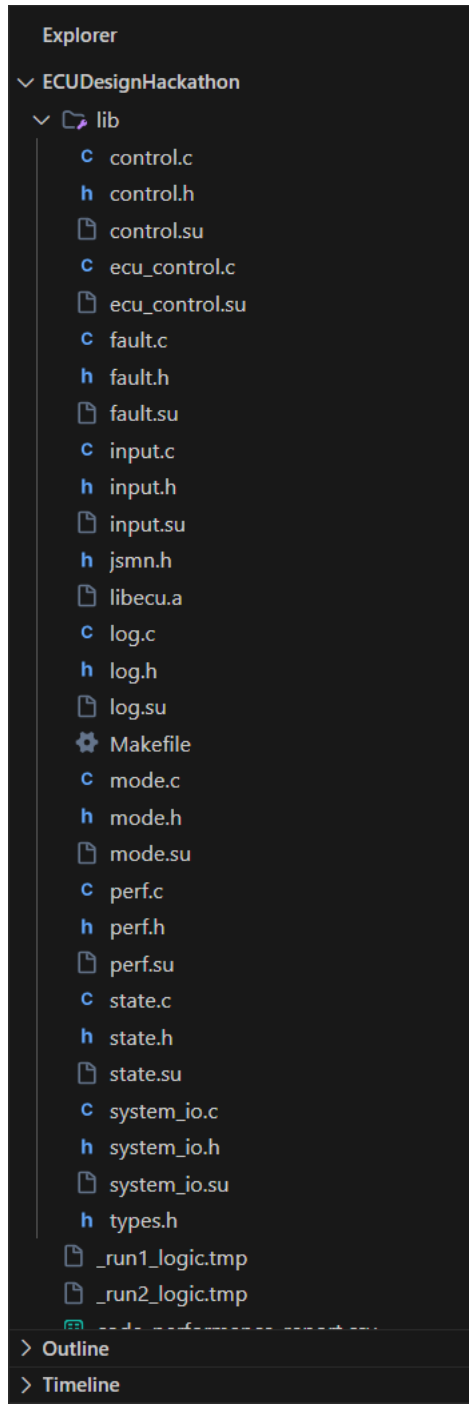
Contents

Folder Structure





The project is organized into a modular structure that separates the Simulation Environment from the ECU Firmware Logic, following best practices for embedded software development.

Root Directory (\)

Contains the application entry point and the artifacts required to run the simulation.

- **main.c:** The host application. It initializes the virtual environment and manages the user menu.
- **test_cases.json:** The persistent test database. All automated simulation scenarios are defined here.
- **ecu_sim.exe:** The final compiled simulator.
- **log.txt / *.csv:** Output files generated during the simulation (logs, functional reports, and performance metrics).

Library Directory (\lib)

This is the core of the project, containing the "firmware" being simulated. It is compiled into a static library (libecu.a).

- **types.h:** The Single Source of Truth. Contains all shared enums (Modes, States) and structures (VehicleStatus).
- **ecu_control.c:** The Scheduler. It handles JSON parsing and coordinates the execution of all other modules.
- **Core Logic Modules:**
    - **input.c**: Sanitization and validation logic.
    - **mode.c:** Ignition state machine.
    - **control.c:** Safety cross-check logic (detection).
    - **fault.c:** Diagnostic persistence and counters.
    - **state.c:** Final system health arbitration.
- **System Wrappers:**
    - **system_io.c:** The Hardware Abstraction Layer (HAL). It wraps standard functions (printf, fopen) to keep the ECU logic portable and testable.
    - **perf.c:** Low-level CPU cycle measurement tools.
- **Makefile:** The build script that compiles the library with size optimizations (-Os).